

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTMPE604D - ROBOTICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Work volume in Cartesian Configuration of Robotics is _____ shape
a) Rectangular b) Cylindrical c) Parallel Sphere d) Irregular
 - 2 Which joint gives linear motion in robots?
a) Revolute b) Spherical c) Prismatic d) Cylindrical
 - 3 The input to inverse kinematics is:
a) End-effector coordinates b) Link dimensions
c) Joint angles d) Gripper force
 - 4 What is the typical input to a robot path control system?
a) Length of link b) Coordinates of start and end positions
c) Tool weight d) Operator's age
 - 5 What is the size of a homogeneous transformation matrix used in robotics?
a) 2×2 b) 3×3 c) 4×4 d) 6×6
 - 6 Which motor provides precise angular positioning without feedback?
a) Servo b) Stepper c) AC d) DC
 - 7 Which drive is most suitable for high payload robotic arms?
a) Pneumatic drive b) Hydraulic drive c) AC drive d) Hybrid drive
 - 8 Optical encoders are mainly used to measure:
a) Vibration b) Rotation angle or speed c) Magnetic field d) Temperature
 - 9 What is the purpose of a vacuum gripper?
a) To grip metallic parts b) To generate heat for welding
c) To hold smooth, flat surfaces using suction d) To measure pressure
 - 10 Which sensor is best suited for detecting touch or contact?
a) Proximity sensor b) Thermocouple
c) Tactile sensor d) Gyroscope
 - 11 AML stands for:
a) Advanced Machine Language b) Automatic Machine Language
c) A Manufacturing Language d) Artificial Motion Language

- 12 Problem representation in AI involves:
- a) Writing robot code b) Defining initial and goal states
 - c) Selecting actuators d) Robot calibration

- Q2. Solve the following.
- A) State and explain various robot configurations in detail. Also write their work envelope and applications. 6
 - B) Write a short note on Robot Control System. 6
- Q3. Solve the following.
- A) Explain homogeneous transformation in Robot for rotation and Translation. 6
 - B) Explain different types of links and joints used in robotics. 6
- Q4. Solve Any Two of the following.
- A) Explain in brief DC Motors used in Robotics, also write its applications in Robotics. 6
 - B) Explain hydraulics drives. What are the key features of hydraulics drives? Write limitations of the same. 6
 - C) Explain in brief the selection of electric actuators. 6
- Q5. Solve Any Two of the following.
- A) Explain with the help of neat sketch any two of the linkage activated robotic grippers 6
 - B) Describe in brief the following (1) Position Sensors (2) Proximity Sensor 6
 - C) Compare Proximity Sensors: Inductive, Capacitive, and Photoelectric. 6
- Q6. Solve Any Two of the following.
- A) Discuss in detail economic analysis of Robots. 6
 - B) Explain in brief various programming languages used in Robotics with its unique features. 6
 - C) Explain in brief the role of robot in Manufacturing industry. 6

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